

MODULE 03

Shoulder MRI: Rotator Cuff, Instability & Arthroplasty

Goutallier · Patte · CSA · Walch · Seebauer · Hamada · Arthroplasty Complications

1.0 AMA PRA Cat 1

Shoulder

CME Article Review

LEARNING OBJECTIVES

1. Apply Goutallier grading (0–4) to rotator cuff muscle fatty infiltration on T1 MRI.
2. Classify supraspinatus retraction using the Patte staging system.
3. Interpret Critical Shoulder Angle (CSA) and its pathoanatomic implications.
4. Classify glenoid morphology using the Walch classification (A1–D).
5. Recognize RCA severity using Seebauer and Hamada classifications.
6. Identify aTSA and rTSA-specific complications: notching, loosening, ALTR.

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1. Goutallier Fatty Infiltration (Parasagittal T1 / CT)

Assessed on parasagittal oblique T1 MRI at the coracoid tip level ("Y-view"). Grade all four cuff muscles separately. Grade ≥ 2 predicts poorer outcomes; Grade ≥ 3 = strong contraindication to isolated repair.

Grade	Fat Content	T1 Appearance	Surgical Implication
0 — Normal	No fat	Homogeneous muscle; no streaks	Excellent repair expected
1 — Minor	Scattered streaks	Fat streaks within muscle	Good outcome; repair indicated
2 — Moderate	Fat < muscle	Fat less than muscle	Moderate; repair still beneficial
3 — Significant	Fat = muscle	Equal fat and muscle tissue	Poor; repair controversial; consider rTSA
4 — Severe	Fat > muscle	Fat exceeds muscle; marked atrophy	Irreversible; primary rTSA

CRITICAL

INFRASPINATUS is the critical muscle. Goutallier ≥ 2 infraspinatus alone predicts repair failure regardless of supraspinatus grade. Always grade all four muscles separately and report each.

2. Patte Retraction, CSA & Walch Glenoid

Patte Classification — Coronal Oblique MRI

Stage	Stump Position	Repair Feasibility
Stage 1	Near greater tuberosity (at insertion)	Easy; primary repair achievable
Stage 2	At level of humeral head	Moderate; may need interval slide
Stage 3	At or beyond glenoid level	Significant; primary repair may not be possible

TEACHING PEARL

Stump remnant >15 mm predicts repairability (Meyer et al. 2012). Report: (1) Patte stage, (2) stump length in mm, (3) Goutallier of supraspinatus AND infraspinatus together.

Walch Glenoid Morphology (Axial MRI/CT)

Type	Description	Surgical Approach
A1	Centered; minimal erosion	Standard glenoid; good fixation
A2	Centered; central vault medialization	Vault bone graft may be needed
B1	Posterior subluxation; no biconcavity	Eccentric reaming or augmented component

Type	Description	Surgical Approach
B2	Posterior subluxation; BICONCAVE; >2.5 mm step-off	Augment or bone graft; complex
B3	Monoconcave posterior; retroversion >15°	Significant correction; grafting often needed
C	Retroversion >25° (dysplastic)	Severe; augmented component or rTSA

3. RCA — Seebauer, Hamada & Arthroplasty Complications

Seebauer RCA Classification

Type	Description	rTSA Risk
IA — Centered, stable	Superior migration; CA arch intact	Favorable
IB — Centered, medialized	Migration + medialization; early bone loss	Acceptable
IIA — Decentered, limited	Anterosuperior escape; acromion eroded	Cautious
IIB — Decentered, destabilized	Anterior escape; AC joint/clavicle eroded	High risk

Hamada RCA Progression

Grade	Findings
1	AHI >6 mm; no articular changes
2	AHI ≤5 mm; no articular changes (superior migration)
3	AHI ≤5 mm + acetabularization of acromion
4a	AHI ≤5 mm + GH joint space narrowing
4b	AHI ≤5 mm + narrowing + acetabularization
5	Humeral head collapse

Sirveaux Scapular Notching (rTSA)

Grade	Extent	Action
1	Within the inferior scapular pillar	Monitor
2	Contacts inferior baseplate screw	At risk — close follow-up
3	Beyond inferior screw; baseplate stability affected	Consider revision
4	Extends to inferior baseplate border	Critical — imminent failure

References

- Goutallier D et al. Clin Orthop 1994.
- Walch G et al. J Arthroplasty 1999.
- Meyer DC et al. Am J Sports Med 2012.